

Benchmark for Crack Re-Identification in Ticket Management Systems

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Abstract—Building ticket management systems must determine whether a newly reported defect corresponds to an existing ticket or requires a new one. Repeated reports of the same crack from different viewpoints fragment history and duplicate work. We study this association as crack re-identification and introduce CRACKID (140 photographs, 17 walls, 212 identities in the frozen benchmark snapshot: 6 validation walls and 11 test walls with 19 matchable test identities). We evaluate ten zero-shot crop-based matchers from four families under retrieval, thresholded decisions, and open-set identification. Retrieval varies (Rank-1 0.352–0.654), but validation-calibrated pairwise decisions stay at or below the best-constant floor (F_1^{ov} 0.293–0.351 against a 0.305 floor; oracle F_1^* 0.305–0.359). Context and crack-erasure controls implicate wall location over crack appearance. A geometric baseline registering wall texture with cracks masked out and matching projected masks reaches 0.768 detection-and-identification rate at 0.1 false alarms on 48% of pairs, against 0.429 for the best crop method, 0.404 for the best context control, and 0.000 for a coverage-only control. All figures are point estimates on a small wall-clustered sample with single-annotator homography-propagated labels, so the open-set advantage is provenance-conditioned and ranking is short-baseline within-visit correspondence. This establishes a dataset and benchmark for the visual association step, leaving cross-visit performance and ticket reduction for future work.

Index Terms—crack re-identification, dataset, benchmark, building maintenance, structural inspection, homography, geometric alignment, annotation quality

I. INTRODUCTION

Building ticket management systems organizes reports of physical defects into records that support inspection, prioritization, intervention, and follow-up. A ticket should maintain the history of one incident. Still, the same crack may be photographed repeatedly by different users or inspections, spawning duplicate tickets that fragment history and duplicate assessment work.

To address this, each new observation must be linked to the existing ticket, or a new ticket must be created for any damage that has not been recorded.

This is a problem of visual re-identification in terms of photographic evidence. There are two cases to consider: (i) the morphology remaining unchanged even when the viewpoint, distance, field of view, illumination or partial visibility varies; and (ii) physical damage that has changed (by extending, widening or branching) yet whose identity must still be retained in the face of such changes, in order to distinguish between damage that has changed and damage that has been

acquired. An error in either of these cases results in a duplicate or in the wrong attribution of the observation.

Current methods for detecting cracks focus on detection and segmentation within individual images. Since datasets like CrackSeg9k [1] and OmniCrack30k [2] do not include cross-observation identities, it is not possible to associate an observation with a ticket. The applicability of person/object re-identification [3], [4] to cracks—given that different defects may look similar and one defect can vary greatly—has not been tested.

We have developed CRACKID in order to determine whether the photographs show the same wall crack: it involves 140 photos of 17 walls taken at one time (6 for validation and 11 for testing; there are 19 test identities that can be matched, with $n_Q=305$). The real photos account for variations in how the images were acquired during a single visit; a synthetic approach is used initially to investigate changes in appearance and growth. Improvement in performance across visits using real photos is something that will have to be left for future work. The accuracy figures given are point estimates based on a small sample of walls clustered together.

The benchmark involves testing ten matchers in a zero-shot manner on CRACKID, with the candidate retrieval being separated from the thresholded decisions regarding whether cracks are the same and with open-set identification involving rejection: merely ranking a record highly does not mean that a report should be attached to it. The use of context together with crack-erased control samples separates the signals indicating cracks from those indicating wall position, and a geometric baseline is used to detect walls and to compare the projected masks.

The study looks at RQ1 (the association under acquisition change), RQ2 (the roles of context and geometric alignment), and RQ3 (behaviour under synthetic appearance/growth change); for RQ1 and RQ2 real single-visit photos are used while RQ3 is an exploratory synthetic study. The aspects of duplicate reduction at the ticket level and cross-visit evaluation will be dealt with in future work.

Contributions: (i) the CRACKID method together with cross-observation identities and information about the source of the annotations (Section III-B); (ii) a benchmark that divides the tasks among retrieval, making same-defect decisions, and open-set identification (Sections IV and VI); (iii) the use of context controls and a geometric reference together with an

analysis of coverage (Sections V, VI-C, and VII-B); (iv) an initial synthetic revisit probe (Section VII-D).

II. RELATED WORK

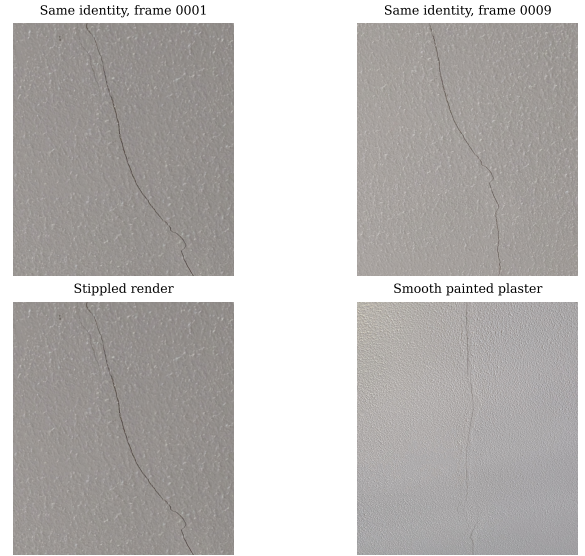
The detection and segmentation of cracks is the focus of crack analysis [5]–[7]; the datasets used in this area also deal with single images and the consolidation studies follow the same scope [1], [2], and multi-view pipelines combine one inspection into a surface or a segmentation [8] without establishing instance correspondence. Structural-health monitoring compares cracks over time with related equipment [9]–[12], but the identity of the cracks is *given by construction* (using a fixed camera and a marked location): instead of measuring a known crack they decide on associations under an open-set protocol, which is our aim.

The setting we use is object re-identification [4] together with open-set identification [13]; keypoint matching methods such as SIFT [14] and ORB [15], learned correspondences (including SuperGlue [16] and LoFTR [17]), and region embeddings [3], [18]–[21] can all be applied to crack detection, with all of them being evaluated here in a zero-shot manner without any CRACKID fine-tuning (for the YOLOv8 row: layer-9 features from `yolov8n.pt` [22]; LightGlue [23] and NetVLAD [24] are left as possible retrieval methods for future work). Vehicle-damage re-identification across claims [25] follows the same open-set protocol as ours; the other work in this area focuses on detection and severity rather than on association [26]–[28]. The process of aligning the images and then reasoning about presence and absence constitutes change detection [29], [30] using classical planar registration [31]–[33] and an ECC fallback [34]; our key point is that the alignment and matching targets should be *disjoint*, the homography being determined by the wall since it is the crack that is to be identified.

III. DATA ACQUISITION

A. Acquisition and composition

The CRACKID frozen benchmark snapshot consists of 140 photographs of 17 exterior walls which were taken at a single location over four half-days using two consumer handheld devices (the data can be obtained through Mendeley Data [35]). The walls have both stippled render (textured) and smooth painted plaster; in the case of the smooth plaster, the only visible features are cracks. The dataset is split into six validation walls (walls 01 to 06; 66 photos, 66 identities) and eleven test walls (walls 07 to 17; 74 photos, 146 identities). Only those identities that appear in at least two photographs are considered to be answerable queries (18 in the validation set and 19 in the test set; $n_Q=305$ in the test set and 439 in the validation set). Additionally, two further walls (walls 18 and 19) have been included and are therefore omitted from all the figures in this paper. Each wall is recorded by a continuous walk-around, with a median time interval of 3 seconds between consecutive frames and a maximum duration of 81 seconds; it is for this reason that the data includes within-visit viewpoint



The images shown here are real CrackID images; the top one shows a single identity spanning non-adjacent frames, while the bottom one shows the stippled render alongside the smooth plaster. The labels were obtained via homography propagation and have not been independently audited.

and scale correspondence, which is regarded as an upper bound on multi-visit performance.

Adjacent-frame retrieval. In the case of the 305 test queries which did have an answer, we use the adjacent frame as the starting point for the retrieval process. As the gaps shown in Section VII (ranging from 305 down to 205, then to 149 and finally to 105 for the gaps numbered 0 to 3) make clear, Rank-1 is still achieved as the number of items decreases. This result shows that the method is robust in the face of short-baseline variations rather than indicating a lack of relevance.

B. Annotation, and why it needs auditing

The ORB homography method operates one-to-one, so for each crack, a single click passes from one frame to the next, and the fragment then combines the segments that constitute the individual physical crack. The greatest risk of an incorrect merge is at this stage, such as changing a hard negative into a free positive. Although the number of merges is provided, their accuracy is not. More precisely, 95 of the 212 identities (45

IV. BENCHMARK PROTOCOL

The benchmark assesses the visual association aspect by determining, for a new crack observation, whether it matches one of the recorded defects or if it does not appear in the reference set. Since annotated identities act as stand-ins for permanent defect records, the measurement is of association accuracy rather than duplicate-ticket reduction or overall workflow performance.

TABLE I
COUNTING UNITS FOR EVERY METRIC (FROZEN SNAPSHOT, $n_Q=305$
TEST).

	Validation	Test
Walls / photographs / identities	6 / 66 / 66	11 / 74 / 146
Identities in ≥ 2 photographs	18	19
Answerable queries	439	305
Positive query-gallery pairs	26 012	5 722
Valid query-gallery pairs	89 118	31 822
Same-wall image pairs	394	301
Scored image pairs (with valid cells)	341	255

A. Task formulation and candidate records

A wall means a single physical wall; an identity is the association between observations relating to the same crack (as explained in Sec. III-B). A query involves a single crack instance; its gallery comprises other instances from photographs of the same wall, provided that the wall is known. Cross-wall search is not included in this approach. A query is regarded as known if its gallery contains the same identity, otherwise it is unknown. Each method assigns scores to the query-gallery pairs in order to carry out ranking, thresholding decisions, and identification with rejection. The unit used in pairs is a query-gallery pair; an image pair is the unit of registration or assignment. The Hungarian one-to-one constraint is applied only in the case of each image pair; more than one observation may refer to the same defect record.

The numbers of the tests and how common they are are presented in Table I for the frozen benchmark snapshot (this covers 6 validation walls, namely walls numbered 01 to 06; 11 test walls, namely walls numbered 07 to 17; the walls labelled wall18 and wall19, which were added later, are omitted from all the figures). Registration is performed for all 301 pairs of test images that are identical in wall; 255 of these result in scored pairs with validity-mask cells (for further details, refer to Sec. VII-B).

B. Unchanged damage under different acquisition conditions

The main evaluation involves carrying out real walk-around inspections on a per-wall basis, with the morphology regarded as essentially unchanged and any differences arising from variations in viewpoint, distance, scale, lighting, and visibility, together with the inclusion of same-wall distractors and unknown queries. The walls are not shared between the validation and test phases. Each method is used exactly as it is without any fine-tuning using CRACKID on shared masks (Sec. V); the thresholds are set using the validation data and then applied to the test data. In this way, within-visit correspondence is established, but neither cross-visit nor progression robustness is addressed.

C. Controlled appearance change and damage progression

A synthetic reanalysis separates brightness variations from those caused by structural growth by using ten photographs from five walls, along with one initial seed, and applying

the same slight affine warp in each case. In condition B, the appearance is degraded (with blur, noise, and changes in contrast or brightness) even though the structure remains unchanged; in condition A, one end of the largest component is extended. The code specifies all parameters. The study compares OSNet with context to the HYBRID-SKEL method (which combines geometric alignment with a fallback/blend based on the skeleton, $\tau=8.0$ px), which differs from the main REG+CHAMFER reference. Because we use only one seed and five walls, the results are preliminary checks, not longitudinal evidence.

D. Decision rules and handling of unavailable scores

When a pair meets the threshold, it is accepted; at the query level, if the threshold is met then only the top entry from the gallery is accepted, otherwise no acceptance occurs; in the case of assignment, the Hungarian algorithm is applied to each image block and any assignments below the validation threshold are discarded. Unscorable pairs are retained: for the pairwise and assignment F1 scores, a predicted negative is used, and in retrieval the pair is given the last rank. Thus, it is possible to distinguish between failing to find a known query and correctly rejecting an unknown query and a supported non-match; since a registration failure is not resolved and may need further examination, it should not be treated as evidence of a new defect. For the assignment F1 score, which equals $2TP$ divided by $(2TP + FP + FN)$, correctly accepted assignments are true positives, incorrectly accepted assignments are false positives, and known instances that are not matched are false negatives.

E. Evaluation metrics and reference baselines

The assessment of known queries makes use of Rank-1, Rank-5 and mAP, whereas pairwise decisions rely on precision, recall and the F_1 measure, with the oracle F_1^* being separated from the validation-calibrated $F_1^{@v}$. The F_1^a is used for assignment. In the case of identification with rejection, DIR (in the case of known accepted-correct cases) and FAR (in the case of unknown accepted cases) are used; the constrained operating point is defined by $DIR@FAR=0.1$, this point differing from the one obtained by means of a fixed threshold. Coverage is the fraction of scored pairs and is provided together with the accuracy. The references include random ranking and the best-constant pairwise F_1 of $2p/(1+p)$ (0.305 at the test prevalence of 0.180; 0.544 on the 0.48 geometric subset; 0.452 at the validation prevalence of 0.292); the full chance levels are given in Table II. Resampling is performed under the shared capture conditions; since there are 11 test walls and 19 matchable identities we provide only point estimates, with wall-level intervals and the release of the score matrix to be considered in future work (Sec. VIII).

V. A GEOMETRIC REFERENCE BASELINE

The baseline test examines whether the location of cracks on the wall is associated with the model: it aligns the wall surfaces to a common frame and then compares the projected

crack masks. It makes the assumption of a roughly planar wall and does not use any CRACKID training. The shared UNet++ masks (which were trained on public crack data only; none of the walls in the training set were from CRACKID) use instance extraction at rates of 200 px and 5 px in order to maintain consistency of the inputs across the different methods (the full segmenter settings are given in the code). The masks are taken as given and their error is not measured.

A. Registration using wall texture

For I_a, I_b , a homography H ($x' \sim Hx$) is estimated from surrounding wall texture with detected cracks dilated and excluded, so alignment is driven by surface features independently of the identity under test. This independence is conditional on the masks and stage, not guaranteed per pair. Full detector settings (SIFT with CLAHE contrast enhancement [36]/ratio/robust estimation) are in code.

B. Registration cascade and validity checks

Background features are generally provided by textured rendering, whereas smooth plaster is not. A three-stage cascade is employed: first, masked SIFT; if too few keypoints remain, then unmasked-SIFT is used as a fallback (this is flagged on a per-pair basis since the crack helps with alignment); and as a final fallback, ECC photometric methods are applied. The candidate images pass plausibility tests (ensuring there are no mirrored or folded quads and that the scale is within the interval $[1/16, 16]$); the stage that survives for each pair is recorded and the inlier RMSE from that stage determines the comparison tolerance. Pairs that are not scored are considered cases of lack of evidence, not evidence that the defects are different.

C. Crack projection and geometric comparison

Accepted registrations project instances from I_a into I_b for comparison by symmetric mean Chamfer distance, which tolerates small misalignments that would collapse IoU on narrow cracks. Let $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2} \left[\frac{1}{|a|} \sum_{p \in a} d(p, b) + \frac{1}{|b|} \sum_{q \in b} d(q, a) \right], \quad (1)$$

The early rejection of distant candidates is considered, and chamfer coverage (the fraction within tolerance τ of the per-pair RMSE, clamped within the range $[2, 16]$ pixels, with the default value at 6 pixels; HYBRID-SKEL uses 8.0 pixels) is assessed separately as its scale-free counterpart (see Sec. VII-C).

D. Correspondence decisions and limits of spatial evidence

For each image pair a one-to-one assignment is made according to the Hungarian method [37] together with a validation threshold (this is the same for all the methods); pairs that are not scored are given a value of $-\infty$ (this is the predicted negative and corresponds to ranking last as described in Sec. IV). An aligned projection onto a wall that is visibly empty is considered a **licensed negative**—that is, evidence of absence which is stronger than

a low embedding similarity—while missed detections, areas outside the frame, and registration failures remain unresolved. The REG+CHAMFER method focuses on damage that is unchanged; the one that deals with progression uses HYBRID-SKEL. The performance depends on surface-dependent failures and on the homography-based annotation provenance.

VI. THE APPEARANCE CEILING

A. Setup

The ten methods are assessed using crack-centred crops and a common validity mask: SIFT and ORB together with RANSAC inliers, SuperGlue and LoFTR using pretrained indoor settings, five embedding-based methods, and re-ID OSNet (using cosine similarity, with all being zero-shot on CRACKID); YOLOv8 makes use of the layer-9 `yolo8n.pt` features. The full input sizes and checkpoints are given in the code repository. For the context controls, OSNet and CLIP are used with a fixed backbone: `@ctx1` increases the size of the instance box by $1 \times$ in each direction, and `@ctx1@nocrack` fills in the crack in the same view in order to isolate the wall context. The REG+CHAMFER method aligns the walls and then compares the projected masks (as described in Sec. V); the COVERAGE-ONLY control assigns a constant score to the registered pairs without examining the cracks. In the synthetic protocol, OSNet with context is compared to the different HYBRID-SKEL configuration. LightGlue and NetVLAD are regarded as future retrieval baselines.

B. Result: appearance buys almost nothing on the decision

The primary benchmark results, as shown in Table II (11 walls; frozen snapshot; $n_Q=305$ answerable test queries), show that the Rank-1 performance varies between 0.370 and 0.613 (ViT-B/16 obtains the highest score; OSNet achieves the best mAP at 0.497; SuperGlue gives the best crop $\text{DIR@FAR} = 0.1$ at 0.384). Yet the oracle F_1^* is in the range from 0.337 to 0.374 and the validation-calibrated $F_1^{\text{@v}}$ lies between 0.331 and 0.338, which shows that good ranking performance does not imply accurate decision-making. The small range of the validation-calibrated F_1 indicates difficulties in transferring a decision threshold across the walls. With 19 matchable test identities, no fine-grained ranking among the crop rows is claimed. All the values reported are point estimates.

C. Is it the crack, or the wall behind it?

Contextual information has a positive effect on both ranking and open-set performance, yet even when it is removed the majority of this benefit is still retained: OSNet's Rank-1 score rises from 0.654 to 0.775 (it reaches 0.711 when the context is removed), DIR@FAR increases from 0.168 to 0.404 (it is 0.393 when the context is removed), and CLIP's open-set performance improves from 0.143 to 0.311 (it amounts to 0.286 when the context is removed). The two views that have the context removed outperform all the methods which use tight crops, indicating that the wall location is the main clue. The pairwise F_1^* remains within a narrow range (from

TABLE II

MAIN RESULT ON CRACKID TEST (11 WALLS; FROZEN SNAPSHOT, $n_Q=305$). F_1^* IS COMPUTED USING AN ORACLE THRESHOLD ON THE TEST SET; ΔF_1^* REPRESENTS THE EXCESS OVER THE FULL-POOL FLOOR. $F_1^{\text{@v}}$ IS CALCULATED USING THE VALIDATION-CALIBRATED THRESHOLD. BOLD FONT INDICATES THE BEST CROP ROW IN EACH COLUMN. THE GEOMETRIC ROW IS BASED ON A SUBSET OF SIZE 0.48 WITH A FLOOR OF 0.544. THEREFORE, ITS ΔF_1^* IS NOT DIRECTLY COMPARABLE TO THAT OF THE CROP ROWS; SEE SEC. VII-B. FOR DETAILS ON SCORING, CHANCE LEVELS, AND COMPUTATIONAL COSTS, REFER TO SEC. IV THROUGH SEC. VII-B. THE METRIC s/QUERY IS DEFINED AS TOTAL_s/n_Q ON A SINGLE WORKSTATION (RELATIVE TIMINGS ONLY).

Method	Family	R@1	mAP	DIR@FAR=0.1	F_1^*	ΔF_1^*	$F_1^{\text{@v}}$	scored	s/query
<i>Reference level (same 0.48 pool as geometry)</i>									
COVERAGE-ONLY	geometry	0.515	0.533	0.000	0.544	0.239	0.544	0.48	—
<i>Crop scope</i>									
ORB [15]	keypoint	0.370	0.372	0.043	0.337	0.032	0.337	1.00	0.077
SIFT [14]	keypoint	0.420	0.408	0.049	0.343	0.038	0.337	1.00	0.155
SuperGlue [16]	learned	0.518	0.434	0.384	0.357	0.052	0.337	1.00	5.196
LoFTR [17]	learned	0.512	0.424	0.285	0.357	0.052	0.337	1.00	7.190
YOLOv8 backbone [22]	embedding	0.525	0.439	0.134	0.374	0.069	0.338	1.00	0.016
ViT-B/16 [18]	embedding	0.613	0.487	0.259	0.344	0.039	0.336	1.00	0.021
DeiT-S [19]	embedding	0.570	0.479	0.223	0.337	0.032	0.337	1.00	0.008
CLIP ViT-B/32 [20]	embedding	0.604	0.471	0.143	0.307	0.002	0.306	1.00	0.050
DINOv2 [21]	embedding	0.567	0.459	0.272	0.337	0.032	0.331	1.00	0.011
OSNet [3]	re-ID	0.603	0.497	0.154	0.338	0.033	0.337	1.00	0.006
<i>Full-image scope</i>									
REG+CHAMFER (reference)	geometry	0.952	0.886	0.768	0.656	—	0.582	0.48	27.92

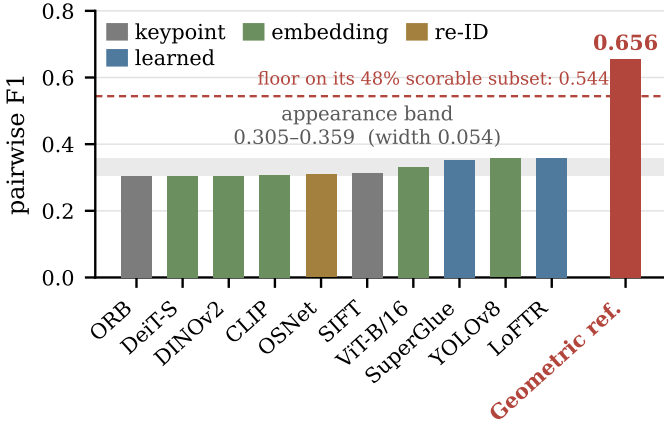


Fig. 1. The appearance ceiling: pairwise F_1^* stays narrow while ranking spreads. Geometry (red) leaves the band but is in the 0.48 pool (own floor 0.544). Table II takes precedence.

0.307 to 0.361) for the base, context, and erased views, while open-set performance improves by +0.236 for OSNet. Nevertheless, although the methods examined are capable of retrieving within-visit observations, reliable automatic identity decisions are still not possible. A ranked list may be of some help to an operator, but acceptance will need more evidence.

VII. RESULTS AND DISCUSSION

A. Main comparison and the open-set gap

Geometric Rank-1 achieves a score of 0.952, whereas the highest score, 0.654 (which is attained by both ViT-B/16 and OSNet), and the mAP of 0.886 as compared to 0.520 (that of OSNet) (the score is 0.48 compared to 1.00). When coverage

is measured in terms of the amount exceeding COVERAGE-ONLY, the ranking improvements remain substantial, even though the open-set figure is one that coverage does not consider: DIR@FAR=0.1 is 0.768 as against 0.429 (SuperGlue), 0.404 (the best result with OSNet+ctx1), and 0.038 for COVERAGE-ONLY itself. For self-similar hairlines, the crop similarities almost always lead to non-rejection, while an aligned empty projection yields a valid negative. The computation does not deal with the discrepancy (LoFTR takes 8.64 seconds whereas OSNet takes 0.05 seconds per query; Reg requires 27.9 seconds per query and 26.0 seconds per image pair; the timings are given only relative to each other). As for RQ2 (the geometric aspect): alignment offers identification capabilities that go beyond what coverage alone can provide, on the condition that success, surface, and provenance are taken into account. All the figures are point estimates based on 19 matchable identities.

B. Partial coverage, gap structure, and failures

Crop methods manage to get 100

C. Ablations and cost

Coverage (scale-free) costs 0.240 mAP and 0.103 Rank-1 versus Chamfer distance, with DIR nearly unchanged (+0.014): the comparison rule moves ranking, not decisions. Disabling crack exclusion leaves ranking/pairwise unchanged but raises DIR by +0.057; masking is retained for evidence independence, not observed accuracy here. ECC adds $\sim +0.03$ Rank-1 / +0.02 mAP with DIR unchanged. Cost is 26.0 s per image pair (7817.3 s over 301 test pairs, i.e. 27.9 s per query; Chamfer-only thereafter), about $3\times$ LoFTR per query and more than two orders above the cheapest embedding; abso-

Protocol	Method	n_q	R@1	mAP	DIR@FAR _{0.1}	$F_1^{\text{@v}}$	F_1^a
A (growth) OSNet+ctx	HYBRID-SKEL	69	0.275	0.401	0.343	0.569	0.646
	69	0.826	0.647	0.739	0.601	0.631	
B (appear.) OSNet+ctx	HYBRID-SKEL	82	0.274	0.403	0.342	0.574	0.640
	82	0.671	0.570	0.663	0.594	0.609	

TABLE III

HYBRID-SKEL VS OSNET+CTX ON SYNTHETIC REVISIT; F_1 AT VALIDATION THRESHOLD. ONE SEED, FIVE WALLS, NO RESAMPLING: CHECKPOINT, NOT STABILITY CLAIM. HYBRID-SKEL $\neq$ TABLE II REG+CHAMFER.

lute timings are workstation-dependent and SuperGlue/LoFTR were timed in a separate run.

D. A synthetic revisit protocol

Ten source photographs taken from five walls (with one seed) are distorted in the same way using either method (A) single-tip growth or (B) photometric corruption and are then compared with the actual images from those same walls (the full procedure is given in the code; the degree of distortion is small, and the p90 tilt is 75°). HYBRID-SKEL differs from REG+CHAMFER as indicated in Table II; consequently, the entries in the rows represent a controlled variation rather than the main configuration.

In the ranking/open-set cases OSNet+ctx does better (the figures being 0.826/0.671 and 0.739/0.663 as compared to 0.275/0.274 and 0.343/0.342), while HYBRID-SKEL achieves a higher assignment F_1^a (0.646/0.640 compared to 0.631/0.609) and the pairwise $F_1^{\text{@v}}$ values are very similar (0.569/0.574 versus 0.601/0.594); no assignment cell came from pure homography alone ($H\%=0.0$). Ordering thus reverses with metric (RQ3): the Table II advantage does not imply multi-visit robustness, and single-seed controlled growth does not establish real progression performance.

VIII. VALIDITY, LIMITATIONS, AND FUTURE WORK

This study provides the first measurable approach to re-identifying cracks within a single visit and sets up a geometric reference standard. Since the results are based on a small sample of wall-clustered data with annotations from a single annotator and only partial registration coverage, each of the findings listed below is accompanied by the specific step that would be needed to verify or build upon them.

- 1) **Scale the system to handle data from multiple walls and multiple visits.** Gather information from more walls, more surface families, and more annotators in order to establish confidence intervals at the wall level and to enable reliable ranking of the methods. Most important of all, include a second capture session carried out at deliberately different distances and lighting conditions so as to turn the within-visit measurement into a true cross-visit benchmark for ticket management backends.
- 2) **Add annotation that does not depend on provenance.** Although each click was verified by a single annotator who also carried out the triage of any unclaimed points,

agreement has not been measured and the geometric advantage is still dependent on provenance. By including a small subset of labels obtained entirely without using homography propagation, together with the release of per-pair score matrices for use in sensitivity analysis and for wall-clustered intervals and threshold-free AP/AUROC, it would be possible to determine how much of the gap in the open-set case is independent of the provenance of the annotation.

- 3) **Close the gap in registration coverage.** The baseline responds to 48 per cent of the pairs (97 out of 301 image pairs; 15,328 out of 31,822 valid cells) and gives a score of zero on smooth-plaster walls. Registration that is adaptive to the surface—this involving the learning of local features, direct photometric alignment, or sparse depth obtained from consumer devices—would move coverage towards complete evaluation and also further separate the A geometry pipeline based on homography-based provenance.
- 4) **Learn from supervision derived from geometry.** The baseline is not a system that has been deployed but rather a reference that is not learned. If the registration pipeline is used to generate pseudo-labels or if a foundation-model backbone is fine-tuned using correspondences derived from homography, the location-based advantage will be retained in the case of pair registration. Assess the ticket lifecycle by defining a set of metrics at the task level—specifically the duplicate rate and the missed-new-defect rate—and then examine the effects of segmentation, re-identification, and the joint assignment of tickets in a closed-loop simulation of actual inspection procedures.

Taken together, these steps convert the current within-visit reference into a deployable A component which can be used for re-identification across multiple visits in order to handle operational tickets.

IX. CONCLUSION

We have developed a dataset that enables the testing of crack re-identification: CrackID, which includes 140 photographs taken from 17 walls and covering 212 identities in the frozen snapshot (6 of these being validation walls and 11 being test walls; walls 18 and 19 were excluded); the source of the annotations (45

The key result of the benchmark is negative and structural in nature: when a decision is required, ten of the zero-shot matchers from four different families achieve a score between +0.000 and +0.054 above the 0.305 best-constant floor (F_1^* ranging from 0.305 to 0.359), and eight out of the ten entries are at or below the floor when assessed at the validation threshold ($F_1^{\text{@v}}$ ranging from 0.293 to 0.351; LoFTR alone scoring +0.046). Determining a person's identity from a gallery is a separate ability to that of making a decision. The problem does not lie merely in the appearance of cracks: a geometric reference baseline that uses wall texture as a reference while the cracks are masked and Chamfer-matches

are projected achieves a $\text{DIR@FAR}=0.1$ of 0.768 compared to 0.429 for the best crop method and 0.404 for the best context method, correctly identifying 48% of the pairs, and a COVERAGE-ONLY control that makes use of that alignment without looking at the crack itself obtains a score of 0.000 on the open-set metric (although it has a pairwise F_1 of 0.544 on its own 0.544 floor).

As far as operation is concerned, the advantage in the open-set scenario results from an aligned pair serving as a *licensed negative*, that is, positive evidence of absence, since a crop embedding can only show low similarity; it is still not clear whether this benefit is independent of homography-based annotation and it must therefore be regarded as depending on the source. Two issues remain to be addressed: carrying out a real second capture session changes the benchmark from one based on viewpoint to one based on cross-visit, and having a second annotator together with a homography-free audit subset would turn the single-annotator triage into a measured agreement rate. The 48

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